

# Anup Rai

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## SUMMARY

Electronics and Communication Engineering student focused on machine learning, data analytics, and LLM-based systems. Hands-on experience building end-to-end intelligent applications, including retrieval-augmented generation (RAG) pipelines, local LLM-powered question answering systems, and real-time computer vision interfaces. Strong foundation in Python, Scikit-learn, and data processing with practical exposure to embedding-based retrieval and context-aware AI systems. IEEE Best Paper Award recipient (AI ML Track, 2026), with a demonstrated ability to translate research into deployable solutions.

## TECHNICAL SKILLS

**Programming Languages:** Python, C, C++

**Machine Learning:** Supervised/Unsupervised Learning, K-Means, Scikit-learn, NumPy, Pandas

**Probabilistic Models:** Bayesian Knowledge Tracing (BKT), Deep Knowledge Tracing (DKT)

**Computer Vision:** OpenCV, MediaPipe

**Tools & Platforms:** Git, Jupyter, Google Colab, Unity, Ollama

**Other:** Graphic Design, Video Editing

## PROJECTS

### Inclusive Mixed Reality Math Experience Using ML-Based Models

2025 – 2026

*Final Year Project*

*Python, OpenCV, MediaPipe, Scikit-learn*

- Built a real-time gesture-controlled tutoring system integrating computer vision and adaptive ML
- Implemented Bayesian Knowledge Tracing (BKT) for adaptive mastery modeling
- Applied clustering techniques to enable personalized learning pathways

### LocalPDF-QA

2026

*LLM Application*

*Python, LLMs, Ollama, Retrieval QA, Streamlit*

- Built a fully local PDF-based question-answering system using open-source LLMs via Ollama
- Implemented document parsing and retrieval pipeline for efficient knowledge extraction

### Local Letter RAG

2026

*LLM / RAG System*

*Python, RAG, LLM*

- Letter ingestion with embedding-based retrieval
- RAG answers grounded in local context

### MediaPipe Cube Control via Hand Gesture

2026

*Computer Vision / HCI*

*Unity, C#, MediaPipe*

- Developed a low-latency real-time hand-tracking interface using MediaPipe integrated with Unity
- Designed modular gesture-state architecture for scalable gesture recognition

## RESEARCH

### DKT-Based AR Framework for Predictive Learning Assessment

2025 – 2026

*IEEE ICTP 2026 | Best Paper Award – AI & ML Track*

- Designed an adaptive Augmented Reality system using Deep Knowledge Tracing (DKT)
- Developed system architecture following Research-Through-Design (RTD) methodology
- Achieved Accuracy: 0.84 and PR-AUC: 0.82 in real-world classroom deployment

## EDUCATION

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**Nepal Engineering College, Pokhara University**  
*B.E. in Electronics & Communication Engineering*  
**Pokhara Aadarsha Secondary School**  
*Science (Grade 11 – 12)*

Bhaktapur, Nepal  
2021 – 2026  
Pokhara, Nepal  
2019 – 2021

## CERTIFICATIONS & HONORS

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- **Best Paper Award** – IEEE ICTP 2026 (AI & ML Track)
- Data Science with Python – Code IT
- IEEE Student Volunteer